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Consideration of environmental impacts in health technology assessment

Jake Williams | 2025 Deeble Scholar
Lancet Countdown Oceania Fellow
PhD Candidate
School of Public Health, Faculty of Health and Medicine, University of Sydney
E: Jake.Williams@sydney.edu.au

Adj AProf Rebecca Haddock
Executive Director | Knowledge Exchange
Deeble Institue for Health Policy Research
Australian Healthcare and Hospitals Association Limited

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Key messages

Australia's health system is a major contributor to national greenhouse gas emissions. Reducing these impacts is a pressing priority. One way to address this, which is gaining international attention, is to consider the environmental impacts of healthcare in health technology assessments (HTA) and decision-making processes.

HTA is complex, and there is no current consensus that environmental impacts should be formally considered within it. Work to determine if this is acceptable in Australia, and how it could be implemented, should be prioritised. This should align with similar work underway by international counterparts.

The range of methods available to include environmental impacts in HTAs have not been thoroughly tested in Australia. The benefits and limitations of proposed methods need to be explored to understand their feasibility, and their impact on the decision-making process.

There is a lack of high-quality data on the environmental impact of health technology products to use as HTA inputs. Investment in Australia's capacity to conduct life cycle assessments of healthcare products and services must be made to address this.

Considering environmental impacts in HTAs could drive meaningful progress to make the healthcare system more environmentally sustainable. Work to understand how this can be done in Australia should be prioritised and enabled.

Recommendations

1. Prioritise environmental considerations in HTA reform: Recommendation 43 of the HTA Review must be prioritised for implementation, backed by clear timelines for assessing feasibility and implementation options.

2. Test methods available to include environmental impacts in HTA: The range of methods available to include environmental impacts in HTA should be tested in the Australian context to understand the benefits, limitations, and feasibility of their use on the decision-making process.

3. Build capacity to generate environmental impact data: Australia's capacity to estimate the greenhouse gas emissions of healthcare products and services should be strengthened. This will require an investment in the skills required to conduct life cycle assessments and for environmental data to be shared and made available.

Table of Contents

Key messages 1
Recommendations 1
Introduction 3
Healthcare contributes to climate change 3
Sustainable healthcare is better for patients and the environment 3
Health technology assessment can contribute to sustainable healthcare 4
Prioritising environmental sustainability in HTA 5
Framework for reform in Australia 5
International counterparts 7
Prioritising effort and the challenges of reform 8
Methods to include environmental impacts in HTA 9
Current Australian guidelines 9
Testing methods for use in Australia 11
Bringing expertise to HTA bodies 12
Generating high-quality environmental data 13
The environmental impact of most healthcare activities is unknown 13
Generating the evidence requires significant investment 15
Establishing a healthcare environmental impact database 16
Conclusion and Recommendations 17
References 19

Introduction

Healthcare contributes to climate change

In 2023, Australia emitted 441 million tonnes of carbon dioxide (CO_2-e) with most of these emissions coming from electricity generation (DCCEEW 2024). In 2022, Australia committed to reduce its emissions by 43% by 2030 and to net zero by 2050 (Climate Change Act 2022). Achieving this will require decarbonisation efforts in all sectors, and the health system is increasingly being recognised as a large source of emissions and an area that can be made more sustainable.

Australia's health system is large, employing one in seven people and contributing about 10% of the nation's gross domestic product (ABS 2022; AIHW 2024). It is a major source of greenhouse gas emissions, estimated to account for 5.4% of our total emissions, or 23.5 Mt CO_2-e (DoHAC 2025a). Although the health system has many other types of environmental impact, most work has focused on greenhouse gas emissions, and it has been argued that these should be the priority of current research efforts to limit human induced climate change (McAlister et al. 2022).

The majority of the health system's emissions come from supply chains for products used in clinical care (Barratt et al. 2022). An assessment of Australia's healthcare emissions found that hospitals are the biggest contributors to emissions (44%), followed by pharmaceuticals (19%), while medical services, community and public health, general practice, and other health practitioners account for an additional 18% (Malik et al. 2018).

Most of these emissions are indirect and not related to energy consumption (Rodríguez-Jiménez et al. 2023). As a result, while the increased use of renewable energy in Australia's energy sector is an essential part of the decarbonisation process, these efforts alone will not bring the healthcare sector to net zero emissions. Achieving this will require healthcare to become more sustainable.

Sustainable healthcare is better for patients and the environment

A sustainable healthcare system has the capacity to deliver care now and in the future (WHO 2017). In the context of climate change, this means ensuring current practices do not unnecessarily emit greenhouse gases that will risk the health of future patients.

Descriptions of sustainable healthcare often refer to a 'triple bottom line' framework, and include considerations of the economic, social, and environmental impacts (Mortimer et al. 2018). Environmentally sustainable healthcare has been defined by five principles (Mortimer 2010):

1. Disease prevention and health promotion
2. Patient education and empowerment
3. Lean service delivery
4. Preferential use of treatment options and medical technologies with lower environmental impact
5. Sustainable operational resource use

These principles aim to prevent illness, provide patient with greater responsibility to manage their health, provide leaner services, use low carbon technologies, and use sustainable practices. Environmental impacts are reduced, and patient outcomes improved, at each step in this process and governments are well placed to enable and drive change in all five areas.

In Australia, the National Health and Climate Strategy vision statement calls for a sustainable, resilient, high-quality, net zero health system (DoHAC 2023). Achieving this will require action in all parts of the health system.

One area of focus is health technology assessment (HTA), where there is interest in understanding how environmental impacts might be included in the decision-making process to ensure low carbon technologies are being selected when appropriate.

Health technology assessment can contribute to sustainable healthcare

HTAs aim to ensure that new health technologies provide health or other benefits at a cost that is acceptable to society (DoHAC 2024b). Although HTAs are not used in every country or health system, they exist in many high-income countries that are most responsible for climate change, including Australia (WHO 2015). HTA methods vary depending on jurisdiction, but typically include considerations of safety, clinical effectiveness, and cost. They do not routinely consider environmental impacts. In Australia's health system, HTA is used to guide decisions for market authorisation and to provide government-subsidised access to new health technologies. HTAs for public subsidy are primarily performed by the Medicare Services Advisory Committee (MSAC) and Pharmaceutical Benefits Advisory Committee (PBAC) and there is growing interest in methods to include environmental impacts in the HTAs conducted by these committees (MSAC 2021; DoHAC 2024b).

HTAs are also conducted by many other Commonwealth, State, and Territory bodies each with their own requirements.

Compared to other industries, the pursuit of sustainable healthcare is unique because trade-offs between environmental impacts and health outcomes may not be acceptable to society or policymakers, given the ethical and clinical imperative to prioritise patient health (WHO 2017; NICE 2023). While sectors like manufacturing or energy may be expected to reduce outputs or profits to meet net zero targets, the health system is currently expected to do so without compromising health outcomes.

Because of this, it is essential that assessments of the environmental impact of healthcare technologies are done alongside the assessment of clinical effectiveness and cost-effectiveness. This is most appropriately done during HTAs.

Prioritising environmental sustainability in HTA

There is growing interest in understanding how to include environmental impacts in HTAs. However, there is no consensus that it should be done, nor is there agreement on the best way of doing it.

In the following sections we argue that, in light of HTA Review Recommendation 43, and the wider interest in this issue, policymakers should prioritise considering the case for incorporating environmental impacts into HTA and how this might best be done.

Framework for reform in Australia

The Department of Health and Aged Care (DoHAC) is considering reform to the HTA process in response Health Technology Assessment Policy and Methods Review (HTA Review) (DoHAC 2024b).

This work commenced in 2025, however changes related to environmental impacts have not yet been publicly prioritised.

The HTA Review was established as part of a strategic agreement between the Commonwealth Government and Medicines Australia (Australian Government, 2003).

The focus of the HTA Review's recommendations are to provide stakeholders and decision makers with tools and processes to address inequalities in access, improve timely access to medicines, improve engagement, and invest in HTA capability to make it adaptable and futureproof (DoHAC 2024b).

Only one of The HTA Review's fifty recommendations concerns environmental impacts. It recommends the Government investigate options to include reporting of environmental impacts in Australian HTA and to consider how these impacts could be used in approval and reimbursement decision (see Box 1).

Box 1 Recommendation 43 of the HTA Review

The Review recommends that the Australian Government, in line with the National Health and Climate Strategy, investigate the following options in consultation with industry and other stakeholders:

a. Reporting of environmental impacts, starting with embodied greenhouse gas emissions, during the assessment of cost-effectiveness by Australian HTA bodies
b. Potential to use this data in approval and reimbursement decisions
c. Potential for public reporting of this data, to inform clinical decision-making
d. The development of guidance documents and examples to facilitate environmental impacts reporting

Source: (DoHAC 2024b)

Box 2 Action 5.1 of the National Health and Climate Strategy

The Australian Government will, in consultation with industry and other relevant stakeholders, review options for including public reporting and consideration of environmental impacts, starting with greenhouse gas emissions, of health technologies, in collaboration and alignment with international best practice in comparable jurisdictions.

Source: (DoHAC 2023)

This will be informed by the National Health and Climate Strategy, which was released in December 2023 (see Box 2) (DoHAC 2023). The Government's response to the HTA Review is being led by the HTA Review Implementation Advisory Group (see Figure 1), which was established in early 2025.

The group's terms of reference state that not all fifty recommendations will be responded to immediately; instead, they will be prioritised by the minister based on the group's advice (DoHAC 2025b).

As a result, the extent to which meaningful policy work progresses in this area will largely depend on the review's prioritisation process, and whether including environmental impacts in HTA is prioritised.

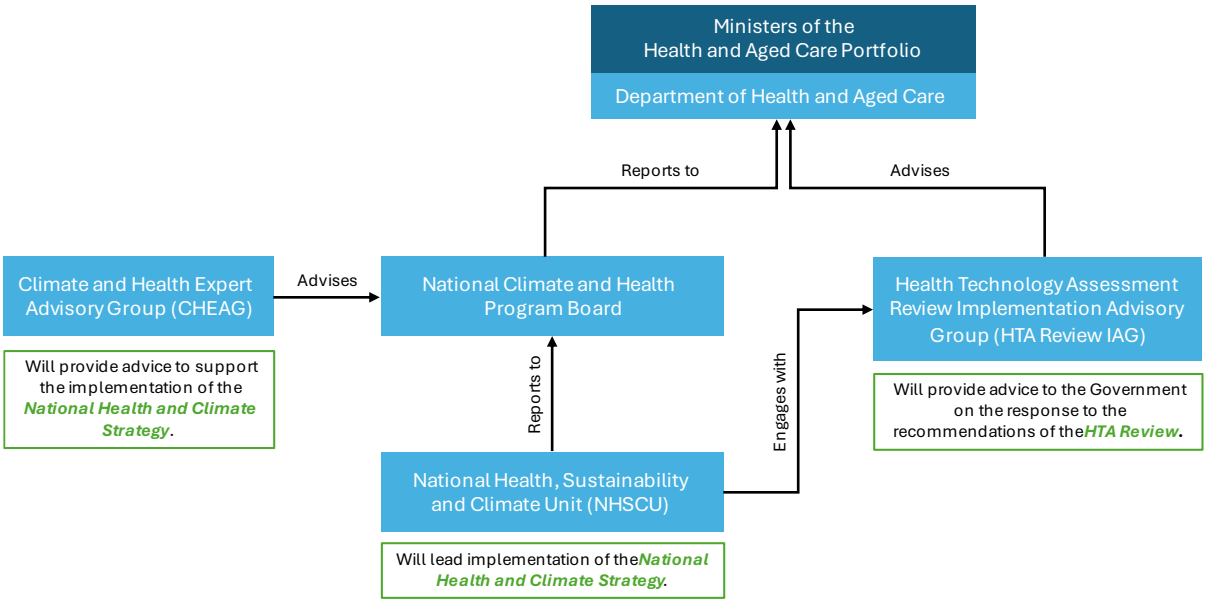


Figure 1: The different bodies responsible for the implementation of the National Health and Climate Strategy and the Health Technology Assessment Policy and Methods Review. Based on information in the National Health and Climate Strategy Implementation Plan 2024-28.

Recognising the importance of HTA in helping to address healthcare sustainability, the National Health, Sustainability and Climate Unit (DoHAC), as part of its work to implement the National Health and Climate Strategy, has identified Action 5.1 on environmental impacts and health technology assessment as a priority for implementation in 2025 (DoHAC 2024a). Aligning the priorities of the HTA Review with the National Health and Climate Strategy is crucial to progress work to explore how environmental impacts might be included in HTAs.

International counterparts

Globally, the effort to include environmental impacts in HTAs is developing rapidly. While there is no final approach that Australia can replicate, progress in some jurisdictions offers potential insights into how Australia can pursue this work (Hodgson et al. 2024).

England

The United Kingdom is a global leader in healthcare sustainability, driven by enforceable and long-standing commitments to reduce greenhouse gas emissions and a healthcare system that is required by legislation to reduce its environmental impact.

In England, this requirement is driven by the Health and Care Act 2022, which requires the National Health Service (NHS) to contribute to net zero targets.

While there is no equivalent legislative requirement in Australia, the National Health and Climate Strategy outlines a commitment to achieving a sustainable, high quality, net zero health system, and identifies a range of actions in pursuit of this, including development of a health system decarbonisation roadmap (Action 4.4) and negotiation of a health system emissions reduction trajectory (Action 4.3).

The main HTA body for the NHS in England is the National Institute for Health and Care Excellence (NICE). Decisions made by NICE determine which services are provided by the NHS.

NICE has committed to developing a method for considering environmental impacts as part of its approval decisions, in support of NHS England's net zero targets (NICE 2025).

This is being informed by work that identified potential options for how NICE could consider environmental impacts (Toolan et al. 2023), and to understand public opinion on doing so (NICE 2023). NICE is currently assessing potential methods and plans to conduct a pilot evaluation once an appropriate method is identified (NICE 2025).

Canada

Canada's HTA body is Canada's Drug Agency (CDA), formerly known as the Canadian Agency for Drugs and Technologies. The CDA has committed to adapting their methods and analyses to include environmental perspectives (CDA 2022). In its 2022-2025 Strategic Plan, the agency outlined its commitment to assess how technologies impact the environmental footprint of health systems (CDA 2022). However, this commitment has not been achieved and implementation details remain unclear (CDA 2019).

The deliberative multi-criteria framework used by CDA's Health Technology Expert Review Panel allows decision-makers to consider the

environmental impact of health technologies when these are reported (CDA 2015), but there is evidence that this is not routinely done (Greenwood Dufour et al. 2022).

Prioritising effort and the challenges of reform

The consideration of environmental impacts as part of Australian HTAs faces a number of challenges, including methodological limitations, data availability, workforce capacity, and operational constraints (DoHAC 2024b; McAlister et al. 2022; Guirado-Fuentes et al. 2023). The feasibility of routinely considering environmental impacts in Australian HTAs will not be known unless work to fully explore these issues is prioritised by government.

The ongoing investment in broader HTA reform provides a timely and appropriate opportunity to deliberate on these issues. This process can ensure that the issues are considered within the broader context of the HTA Review. Australia is uniquely placed to become a global leader in the effort to identify a suitable role for consideration of environmental impacts in HTA.

Box 3 HTAs are not always conducted

HTAs play an important role in determining whether new technologies are effective and cost-effective. In situations where an HTA is going to be undertaken it may be possible for environmental impacts to be considered as part of that process. However, HTAs are not undertaken for all products.

Previously approved technologies

For both MSAC and PBAC, technologies approved for listing are not routinely re-assessed. Although mechanisms for re-assessment exist, they are not mandatory. Most of the thousands of pharmaceuticals and medical services already listed on the PBS and MBS will not be subject to HTA again under the current system, leaving most technologies out of scope.

Generic medicines

For PBAC, new medicines undergo a full HTA as part of the PBS listing process. However, once a patent expires, generic versions are not re-assessed in the same way as the original branded medicine. Although generics are chemically equivalent, they often have distinct supply chains and manufacturing processes, which can result in significantly different environmental impacts. While it may be possible to assess environmental impacts for new brand-name drugs, this gap leaves most generic medicines out of scope.

Recommendation 18 of the HTA Review, which recommends the Australian Government update post-review frameworks for evaluation, may assist in addressing the limitations of current HTA processes to consider the environmental impacts of previously approved technologies and generics. Consideration should be given to the role environmental impacts could play in the post-listing review of health technology products.

Methods to include environmental impacts in HTA

Methods to include environmental impacts in HTAs have not been thoroughly explored or tested in Australia. We argue dedicated consideration of these options should be prioritised by government as part of the work to implement the HTA Review.

Current Australian guidelines

In Australia, the Medical Services Advisory Committee (MSAC) permits the inclusion of information on environmental impacts in submissions. However, the guidance on providing this information is vague and there is no evidence that this information is currently being provided (Box 4).

This issue partially arises from the absence of clear guidelines on how applicants should report environmental impacts and how the committee should evaluate them.

The Pharmaceutical Benefits Advisory Committee (PBAC) does not have the same allowance for the consideration of environmental impacts in their guidelines (PBAC 2016).

As a result, while non-pharmaceutical applicants may submit environmental information, pharmaceutical applicants may not, in spite of the fact that medicines account for a large portion of health systems emissions (Malik et al. 2018).

To ensure a consistent approach that allows for the inclusion of environmental impacts in submissions, MSAC and PBAC guidelines should be aligned. This should be a priority for the next review of the PBAC Guidelines. Similar work should be undertaken to align the guidelines of other HTA bodies.

Potential methods

Work to describe methods to include environmental impacts in HTAs is ongoing and should be tested in the Australian context to fully understand their feasibility.

Available methods have been the subject of recent review studies and in the following sections we outline available methods and highlight some issues associated with each (Marsh et al. 2016; Desterbecq and Tubeuf 2023; Guirado-Fuentes et al. 2023; McAlister et al. 2022; Pinho-Gomes et al. 2022; Toolan et al. 2023; Williams et al. 2024).

Box 4 Environmental considerations for submissions to MSAC

TG 29.6 Environmental aspects

If there are particular concerns regarding the environmental impact of the proposed health technology, or a key benefit in the way the proposed technology reduces the environmental impact compared with the comparator, these can be outlined (e.g., if there is a reduction in the amount of radioactive waste generated, or a reduction in emissions related to transportation or the manufacturing process).

Source: (MSAC 2021)

Methods have been categorised into four approaches:

- information conduit
- integrated evaluation
- parallel evaluation
- environment focused evaluation.

These approaches were defined by researchers at NICE in the United Kingdom (Toolan et al. 2023).

Information conduit

An information conduit approach proposes HTA bodies collect and publish data on the environmental impacts of a health technology without using these data in the assessment or decision-making process (Toolan et al. 2023). This data would be made publicly available, allowing clinicians, patients, carers, and other stakeholders to incorporate environmental considerations into their own decision-making.

The advantage of this approach is that it makes data about environmental impacts available to clinicians and patients, so they can use this information to inform clinical decision-making (Dunsmore 2023; Batcup 2023).

Many clinician groups, as well as some patient groups, have indicated strong interest in having access to this information, especially in cases where a range of largely clinically equivalent options are available.

The limitation of this approach is that it does not ensure that low impact technologies are preferentially used. Instead, it leaves those decisions to other parts the health system. While this could incentivise the generation of much-needed data, particularly if reporting were made mandatory, it may be an inefficient way of evaluating environmental impacts.

Integrated evaluation

An integrated evaluation approach proposes that environmental impacts be fully integrated into a single quantitative analysis with other health and financial data (Toolan et al. 2023). There are two different ways of doing this that have been proposed and demonstrated: by valuating impacts in dollars and including them on the cost side of an evaluation, or by valuing impacts in health units and including them on the outcomes side of an evaluation (Williams et al. 2024). Most work has focused on the former.

Public opinion indicates support for low environmental impact technologies, provided they do not compromise health outcomes or increase costs (NICE 2023).

While some level of trade-off between environmental impacts and cost might be acceptable, evidence from the United Kingdom suggests that trade-offs between environmental impacts and health outcomes are may not be acceptable to the public (NICE 2023). Integrating environmental impacts directly into a single quantitative analysis makes these trade-offs difficult to manage.

The integrated evaluation approach also relies on the conversion of environmental impacts into other units such as dollars. There is evidence that financial conversions, such as the social cost of carbon, may not fully capture the true cost of environmental impacts and is often undervalued (Rennert et al. 2022).

Finally, the conversion of environmental impacts into monetary units, based on existing literature valuations, may be unlikely to influence the evaluation outcome.

This is due to the relatively low monetary value of greenhouse gas emissions when compared to the overall health benefits and costs of the intervention. For example, a micro-costing study of a surgical intervention in France (Pugliesi et al. 2025) demonstrated that when environmental impacts were converted to dollars, they represented only 3.21% of the total intervention cost. A similar cost minimisation analysis in Australia found that monetised emissions account for less than 3% of total cost (Dunbar et al. 2025).

Parallel evaluation

A parallel evaluation approach involves including environmental impacts in value judgements alongside established methods for evaluating clinical and cost outcomes (Toolan et al. 2023).

In this approach, health economic methods would remain unchanged and environmental impact information would be used to supplement the decision process.

The main methods described to achieve this are to report trade-offs explicitly, to allow the decision-making body to freely consider impacts, with or without structured guidance, or to make value judgements using structured methods such as multi-criteria decision analysis (Williams et al. 2024).

However, this approach places the burden of interpreting environmental outcomes on stakeholders undertaking the HTA, who do not historically have the environmental expertise or experience to do so. While some international studies have explored the use of multi-criteria decision analysis in HTA (Oliveira et al. 2019; Thokala & Duenas 2012), it is not currently used by HTA bodies in Australia. Parallel methods have the advantage of giving decision making bodies greater control over how environmental impacts are factored into evaluations and could avoid unacceptable trade-offs with health outcomes that are introduced with other methods.

Environment focused evaluation

Environment focused evaluation is a proposed approach for HTA bodies to conduct evaluations of health technologies that are expected to result in no change in costs or health gains relative to a comparator (Toolan et al. 2023). This method would require the development of specific decision-making frameworks to allow HTA bodies to consider environmental impacts as the primary outcome to evaluate. In the Australian context this has some precedent with cost-minimisation studies that demonstrate cost savings and no changes to health outcomes (MSAC 2021; PBAC 2016).

Testing methods for use in Australia

The current MSAC guidelines allow for the submission of environmental impacts (MSAC 2021) that potentially align with either of the four evaluation approaches described. In contrast, the HTA Review adopts a more specific perspective (DoHAC 2024b), suggesting that environmental impacts be included in the assessment of cost-effectiveness, aligning with either the parallel or integrated evaluation approaches.

Each of the four proposed approaches, and their challenges, need to be understood in the Australian context before any of them can be adopted. Additionally, while there are a number of evaluations published globally (Williams et al. 2024), there are very few available for interventions in Australia. Case

studies using these methods are urgently needed to determine the most feasible options for the MSAC and PBAC.

Bringing expertise to HTA bodies

Including environmental impacts in health technology assessment (HTA) will require subject expertise within decision-making bodies. For example, MSAC and PBAC members typically have backgrounds in clinical practice, health economics, or consumer advocacy rather than environmental science (DoHAC, 2024, 2025a). Currently, there are no members with environmental science expertise.

In addition, existing members with clinical and economic backgrounds will need to develop their carbon literacy to assess environmental impacts effectively and understand the implications of incremental greenhouse gas emissions (Howell 2018).

Expanding MSAC and PBAC to include environmental experts must be carefully timed. Appointing new members before standard methods are established would be inefficient. Any changes should align with the development of frameworks for incorporating environmental impacts into HTA to ensure expertise is introduced when it is most needed. This is particularly important given the existing shortage of professionals with these skills in the healthcare sector.

Generating high-quality environmental data

There is a lack of high-quality data on the environmental impact of healthcare products and services. Consequently, estimating the environmental impact of a technology under review is challenging, expensive, and time consuming. A broad investment in Australia's capacity to conduct life cycle assessments (LCAs) of healthcare products and services must be made to address this.

The environmental impact of most healthcare activities is unknown

The environmental impact of most activities in the Australian healthcare system is not known. While there has been some work to generate this knowledge, only a fraction of the thousands of routinely-used healthcare products and services have been investigated (Drew & Rizan 2025).

This lack of data prevents environmental impacts from being included in HTAs and the economic evaluations they rely on.

The most widely accepted method to estimate the environmental impact of a healthcare product or service is through a life cycle assessment (LCA). LCAs estimate the impacts across the entire life cycle of the technology, from raw material extraction, material processing, part manufacturing, assembly, use, and end-of-life.

There are two main types of LCA, 1) process-based LCA and 2) environmentally extended input-output analysis.

While each of these methods have their place, it is generally accepted that process-based LCA is the most appropriate way to estimate impacts of discreet healthcare products and services (McAlister, Barratt, et al. 2024), while input output analysis is more appropriate for estimating system-level impacts (McAlister, Barratt, et al. 2024; Malik et al. 2018).

To date, only a small number of LCA estimates for some Australian equipment, medications, tests, procedures, and pathways have been performed (see Table 1).

The scope of the environmental impacts included in a HTA should match the scope of resources included in the cost analysis as closely as possible.

This is generally determined by the perspective of the economic evaluation and the different resources included (Drummond et al. 2015).

As an example, an economic evaluation of alternative radiotherapy treatments might include the cost of resources from radiotherapy, hospital bed days, out-patient hospital visits, hospital overheads, general practitioner visits, nurse visits, ambulance or patient transport use, and social worker visits. If the analysis took a societal perspective, it might also include patient time, carer time, and out-of-pocket expenses (Drummond et al. 2015).[1] To include environmental impacts in this economic evaluation, estimates for the environmental impact of each resource are required, most of which are currently unavailable and would require significant investment to generate.

[1] Chapter 7.4 of Methods for the Economic Evaluation of Health Care Programmes (Fourth Edition) by Drummond et al. (2015)

Table 1: LCA for healthcare products and services in Australia (as of 2025), as included in the HealthcareLCA database (www.healthcarelca.com/database).

Category	Description	Source
Equipment	Anaesthetic drug tray	(Lightfoot et al. 2024)
	Cytoscope	(Wombwell et al. 2024)
	Dental restorative materials	(Smith et al. 2023)
	Flexible ureteroscopes	(Davis et al. 2018)
	Anaesthetic equipment	(McGain et al. 2017)
	Central venous catheter insertion kit	(McGain et al. 2012)
	Anaesthetic drug trays	(McGain et al. 2010)
Medication	Paracetamol	(Davies et al. 2024)
	Anaesthesia for total knee joint replacement	(McGain et al. 2021)
	Anaesthesia machine fresh gas flow	(Zhong et al. 2020)
	Morphine infusion bag	(McAlister et al. 2016)
Tests	Diagnostic imaging	(McAlister et al. 2022)
	Blood tests and urinalysis	(McAlister et al. 2021)
	Blood tests	(McAlister et al. 2020)
Procedures	Total knee replacement	(McGain et al. 2024)
	Coronary artery disease treatment	(Sack et al. 2024)
	Peritoneal dialysis	(McAlister et al. 2024)
	Cataract surgery	(Hong et al. 2023)
	Skin excision	(Tan and Lim 2021)05.06.2025 11:19:00
Pathways	Management of septic shock	(McGain et al. 2018)
Systems	Western Australia's public health system	(Irwin et al. 2024)
	Western NSW Local Health District	(Malik et al. 2024)
	Clinical trial	(Griffiths et al. 2024)
	NSW health system	(Malik et al. 2021)
	Haemodialysis unit	(Lim et al. 2013)
	Ambulance services	(Brown et al. 2012)

Source: (Drew and Rizan 2025)

Generating the evidence requires significant investment

Conducting a process-based LCA requires access to specialist environmental expertise, appropriate software, and considerable time and financial resources (Box 5).

However, once the environmental impact of a resource is estimated, it does not need to be re-estimated for work in the same context, potentially saving future resources (McAlister et al. 2022).

For example, the environmental impact of a packet of paracetamol or aspirin would only need to be estimated once and could then be used by all future economic evaluations requiring an estimate of the environmental impact of that resource in the same context. Efforts to generate this data should prioritise the most common types of healthcare resources and those with the greatest environmental impact.

Although the initial wok to compile this data is likely to be substantial, each estimate can contribute to an inventory that future HTA applicants can use (McAlister et al. 2022).

Perhaps the greatest barrier to compiling this repository of data is that the health and research sector workforce currently lacks the necessary skills to undertake process-based life cycle assessments at the scale needed.

While data is urgently needed to progress research in this area—particularly to generate case studies relevant to the Australian context—the volume of data required for the routine consideration of environmental impacts in HTA is substantial.

This level of data is only likely to be generated if regulatory changes are introduced that require environmental impacts to be considered in HTA.

Box 5 Healthcare Carbon Lab at the University of Melbourne

The Healthcare Carbon Lab has been established at the University of Melbourne with the aim of building a comprehensive inventory of the environmental impact of medical equipment.

It is funded by an award from the University of Melbourne Faculty of Medicine, Dentistry, and Health Sciences Deanery.

The Healthcare Carbon Lab is the only research group in Australia dedicated to generating the environmental impact data necessary to include environmental impacts in HTA.

Source: (University of Melbourne 2024)

Establishing a healthcare environmental impact database

Without an accessible repository of environmental impact data for healthcare, HTA applicants and health economists would need to independently perform an LCA for every resource associated with a technology, and this is impractical.

HealthcareLCA (Box 6) is an international database of LCAs of healthcare products and services and is used to identify inputs for economic evaluations. However, the database does not include detailed information of the processes contained within each LCA and is only a repository of the results.

This means that while it is a useful resource for identifying the environmental impacts of different healthcare products, the data can't be easily integrated into new process-based LCA models.

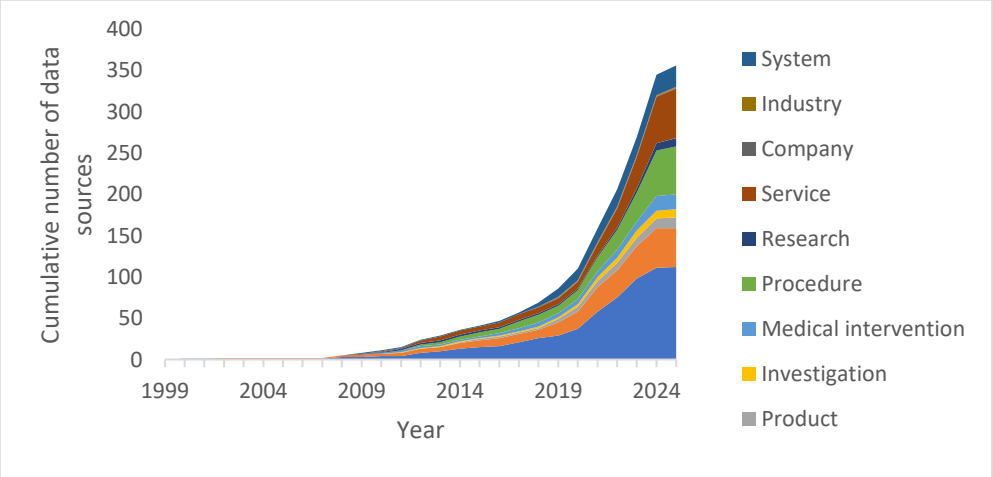
A database is also needed that can integrate directly into LCA models and be used alongside other existing LCA databases such as Ecoinvent and the Australian Life Cycle Inventory (AusLCI). This would allow LCAs of healthcare services to be generated more seamlessly with existing data.

An Australian Environmental Impact Inventory would simplify future assessments and help to reduce duplication within the system. However, to be effective, this resource should be free and publicly accessible, allowing researchers to both use and contribute to the data.

Box 6 HealthcareLCA

HealthcareLCA is a free online database of published LCAs of healthcare products and services, supported by CASCADES, a pan-Canadian initiative to implement sustainable healthcare practices and policies in Canada.

As March 2025 the database contains 358 sources of information for life cycles assessments of healthcare globally.



Source: (Drew and Rizan 2025)

Conclusion and Recommendations

Australia's health system has significant environmental impacts and these need to be reduced.

While healthcare decarbonisation faces unique challenges relative to decarbonisation in other sectors of the economy, such as the central role played by clinical decision making and the acceptability of trade-offs, these should not make it exempt from the shared need to reduce emissions as part of the work to achieve a net zero economy.

We argue HTA can play an important role in making healthcare more environmentally sustainable and that this role can be realised by:

(1) prioritising environmental considerations in HTA reform
(2) testing methods in the Australian context
(3) building capacity to estimate the environmental impact of healthcare products

Recommendation 1: Prioritise environmental considerations in HTA reform

The HTA Review Implementation Advisory Group should prioritise environmental considerations for the HTA Policy and Methods and Review (HTA Review). This will ensure that methods to include environmental impacts in health technology assessment are thoroughly explored, aligning with the goals of the National Health and Climate Strategy and international counterparts.

To achieve this:

- Recommendation 43 of the HTA Review, which calls for exploring how to include environmental considerations in HTAs, should be prioritised for implementation, with clear timelines for assessing feasibility and implementation options.
- A targeted consultation process with industry, researchers, and healthcare providers should be established by government to inform the HTA Review Implementation Advisory Group's response to Recommendation 43 of the HTA Review.

Recommendation 2: Test methods available to include environmental impacts in HTA

Methods to include environmental impacts in HTA, such as those discussed in this paper, should be assessed with a view to considering their potential merits and inclusions in HTA.

This will enable HTA bodies to issue clear guidance on the best way for environmental impacts to be considered. PBAC and MSAC should adopt the same approach to ensure the environmental impact of technologies are assessed uniformly across Australian jurisdictions.

To achieve this:

- Case studies should be developed in collaboration between industry, researchers, and HTA bodies to demonstrate the feasibility and acceptability of methods to include environmental impacts HTAs
- PBAC guidelines should be updated during the next scheduled review to be made consistent with the guidance on environmental considerations for submissions to MSAC.
- Guidance should be developed to describe standard methods for including environmental impacts in MSAC and PBAC submissions and how they are assessed and included in the decision-making process, consistent with the recommendations and outcomes of the HTA Review and National Health and Climate Strategy.
- Members of MSAC, PBAC and their relevant sub-committees should be provided training to ensure they can appropriately interpret environmental impact data. Additional members with environmental expertise should, in due course, be added to the committees.

Recommendation 3: Build capacity to generate environmental impact data for healthcare

Australia's capacity to conduct LCAs of healthcare products and services should be strengthened. This will require an investment in the skills required to conduct LCAs and for data to made available to other researchers.

To achieve this:

- The Australian Government should consider targeted funding through the Medical Research Future Fund to support research to conduct LCAs of key healthcare products and services, prioritising high-impact areas and the most used services.
- A publicly accessible repository for data on the environmental impact of healthcare products must be established, aligning with international standards. This should be developed in partnership between industry and researchers and integrate into existing environmental databases such as Ecoinvent.

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Contact:
Adj AProf Rebecca Haddock
Executive Director | Knowedge Exchange
Australian Healthcare and Hospitals Association.
Email: rhaddock@ahha.asn.au

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